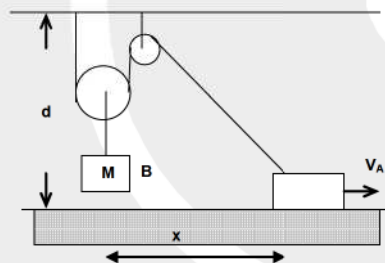




[SINGLE CORRECT CHOICE TYPE]

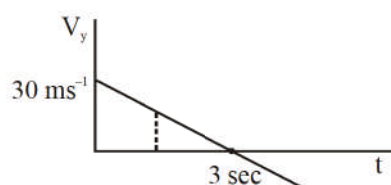
- Q.1 A body starts from rest with uniform acceleration. Its velocity after $2n$ second is v_0 . The displacement of the body in last n seconds is [3]
 (A) $\frac{v_0(2n-3)}{6}$ (B) $\frac{v_0}{4n}(2n-1)$ (C) $\frac{3v_0n}{4}$ (D) $\frac{3v_0n}{2}$
- Q.2 A boy throws an object horizontally with a velocity of 6 m/sec through the window of a bus moving with a velocity of 36 km/h . If the height of the window from the ground is 3 m and the object is thrown normal to the motion of the bus what is the velocity of impact on the ground [3]
 (A) $4\sqrt{6} \text{ m/sec}$. (B) $4/\sqrt{10} \text{ m/sec}$. (C) $4\sqrt{2} \text{ m/sec}$ (D) 14 m/sec .
- Q.3 A body projected with a certain velocity has horizontal range three times the maximum height. Find the change in the angle of projection required to make the horizontal range four times the maximum height [3]
 (A) -8° (B) 8° (C) 16° (D) -16°
- Q.4 A particle projected from the foot of an inclined plane of angle β strikes the plane at a point and rises vertically upwards. If the angle of projection is α find its angle of incidence on the plane (Take elastic collision) [3]
 (A) β (B) $\alpha - \beta$ (C) $\alpha + \beta$ (D) $\beta / 2$
- Q.5 A tractor A is used to hoist the body B with pull by arrangement shown in fig. If A has a forward velocity V_A , find the velocity of the body B [3]



- (A) $\frac{x V_A}{2}$ (B) $\frac{x V_A}{d}$ (C) $\frac{x V_A}{2\sqrt{x^2 + d^2}}$ (D) none of these
- Q.6 The position of a particle along x-axis is given by $x = 1 + t - t^2$. The distance travelled in 2s is [3]
 (A) 1m (B) 2m (C) 2.5 m (D) 3m
- Q.7 A balloon is ascending at the rate of 14 ms^{-1} at a height of 98 m above ground where a packet is dropped, time after which the packet reach the ground is [3]
 (A) 6s (B) 3s (C) 9s (D) 4.9 s
- Q.8 A man wants to reach a point on the other bank of a river that is directly opposite to his starting point. River flows at the rate of 4 km/hr . and swimming speed of the man relative to water is 3 km/hr . Then [3]
 (A) To reach the desired point he should start swimming at an angle of $\cos^{-1}(3/4)$ with stream
 (B) To reach the desired point he should start swimming at an angle of $\cos^{-1}(-3/4)$ with stream
 (C) To reach the desired point he should start swimming at an angle of $\cos^{-1}(4/3)$ with stream
 (D) He can not reach the desired point directly by swimming.



- Q.9 A particle is projected vertically upwards with a velocity of 20 m/sec. Find the time at which the distance travelled is twice the magnitude of displacement [3]
 (A) $2 + \sqrt{4/3}$ sec (B) 1 sec.
 (C) $2 + \sqrt{3/4}$ (D) 3 sec
- Q.10 Two men who can swim with a speed v_1 in still water start from the middle of a river of width d and move in opposite directions always swimming at an angle θ with the banks. What is the distance between them along the river when they reach the opposite banks, if the velocity of the river is v_2 [3]
 (A) $\frac{dv_1}{dv_2} \cot \theta$ (B) $\frac{dv_1 \cos \theta}{v_1 + v_2}$ (C) $\frac{dv_2}{v_1} \tan \theta$ (D) $d \cot \theta$
- Q.11 A particle of mass M moves horizontally in a medium where the magnitude of resistance force is kMV (where k is a constant) and v is the velocity at time t , with the initial velocity u , what is the maximum distance of the particle from the starting point? [3]
 (A) $9/k$ (B) $34/k$ (C) $24/k$ (D) none of these
- Q.12 Two particles are simultaneously projected in opposite directions horizontally from a given point in space where gravity ' g ' is uniform. If u_1 and u_2 were their initial speeds then the time after which their velocities are mutually perpendicular is given by [3]
 (A) $\frac{\sqrt{u_1^2 + u_2^2}}{g}$ (B) $\frac{\sqrt{u_1(u_1 + u_2)}}{g}$ (C) $\frac{\sqrt{u_1 u_2}}{g}$ (D) $\frac{\sqrt{u_2(u_1 + u_2)}}{g}$
- Q.13 A bird flies for 4 s with a velocity of $|t - 2|$ m/s in a straight line, where t = time in seconds. It covers a distance of [3]
 (A) 2 m (B) 4 m (C) 6 m (D) 8 m
- Q.14 A particle is falling freely under the gravity over a short distance from P to Q. M is the mid point of PQ. Then [3]
 (A) its kinetic energy at Q will be double than at M
 (B) the time of motion from P to Q will be less than 1.5 times the time of the motion from P to M
 (C) its velocity at Q will be double its velocity at M
 (D) its average velocity from P to Q is double its average velocity from P to M
- Q.15 Between the two station a train accelerates uniformly at first, then moves with constant velocity and finally retards uniformly. If the ratio of the time taken be 1:8:1 and the max. speed attained be 60 km/h then what is the average speed over the whole journey? [3]
 (A) 48 km/h (B) 50 km/h
 (C) 54 km/h (D) 56 km/h
- Q.16 Figure shows $V_y - t$ graph of a particle projected from the ground. Then its height at $t = 2$ sec will be [3]



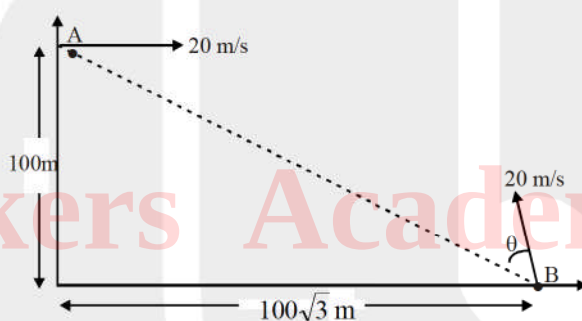
- (A) 50 m (B) 40 m (C) 30 m (D) 10 m

[MULTIPLE CORRECT CHOICE TYPE]

- Q.17 A boat is traveling due east at 12 m/s. A flag on the boat flaps at 53° N of W. Another flag on the shore flaps due north. [4]
 (A) Speed of wind with respect to ground is 16 m/s
 (B) Speed of wind with respect to ground is 20 m/s
 (C) Speed of wind with respect to boat is 20 m/s
 (D) Speed of wind with respect to boat is 16 m/s

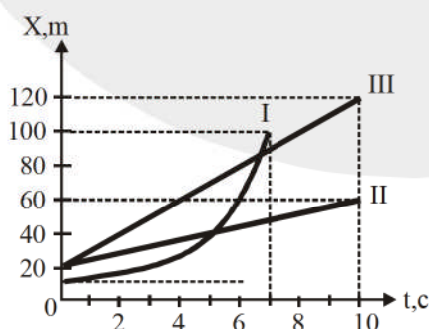
- Q.18 A particle is moving along X-axis whose position is given by $x = 4 - 9t + \frac{t^3}{3}$. Mark the correct statement(s) in relation to its motion : [4]
 (A) Direction of motion is not changing at any of the instants
 (B) Direction of motion changes at $t = 3$ s
 (C) For $0 < t < 3$ s, the particle is slowing down
 (D) For $0 < t < 3$ s, the particle is speeding up

- Q.19 Two small balls A and B are launched in the same vertical plane simultaneously, with same speed of 20 m/s at $t = 0$. Ball A has an initial horizontal velocity and ball B has initial velocity at an angle θ above the line joining A and B as shown. If the projectiles collide in mid-air at time t : [4]



- (A) $\theta = 30^\circ$ (B) $\theta = 60^\circ$ (C) $t = \frac{20}{\sqrt{3}}$ s (D) $t = \frac{10}{\sqrt{3}}$ s

- Q.20 Figure shows the dependence of the coordinates of the body moving in a straight line on time for three cases: I, II, III. Enter the law of motion and construct the graphs of velocity and acceleration versus time in every case. Curve I is a parabola [4]

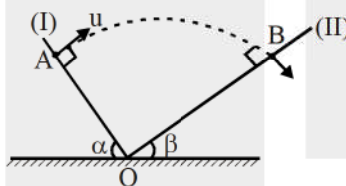


- (A) for I (B) for I (C) for II (D) for III

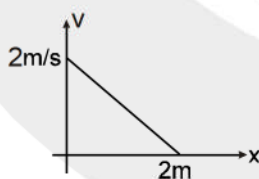
- Q.21 Two particles were projected one by one with the same initial velocity from the same point on level ground. They follow the same parabolic trajectory and are found to be in the same horizontal level, separated by a distance of 1m, 2 seconds after the second particle was projected. Assume that the horizontal component of their velocities is 0.5 m/s. [4]

Which of the following statements will be true about their motion ?

- (A) The horizontal range of the parabolic path is 3 m.
 (B) The maximum height for the parabolic path is 45 m.
 (C) The total time of flight in the parabolic path for each particle = 4s
 (D) The horizontal range of the parabolic path is 6 m.
- Q.22 Two inclined planes (I) and (II) have inclination α and β respectively with horizontal, (where, $\alpha + \beta = 90^\circ$) intersect each other at point O as shown in figure. A particle is projected from point A with velocity u along a direction perpendicular to plane (I). If the particle strikes (II) perpendicularly at B, then : [4]



- (A) time of flight = $u/g \sin \beta$
 (B) time of flight = $u/g \sin \alpha$
 (C) distance OB = $u^2/2g \sin \beta$
 (D) distance OB = $u^2/2g \sin \alpha$
- Q.23 A man is standing on a road and observes that rain is falling at angle 45° with the vertical. The man starts running on the road with constant acceleration 0.5 m/s^2 . After a certain time from the start of the motion, it appears to him that rain is still falling at angle 45° with the vertical, with speed $2\sqrt{2} \text{ m/s}$. Motion of the man is in the same vertical plane in which the rain is falling. Then which of the following statement(s) are true : [4]
- (A) It is not possible
 (B) Speed of the rain relative to the ground is 2 m/s
 (C) Speed of the man when he finds rain to be falling at angle 45° with the vertical, is 4 m/s.
 (D) The man has travelled a distance 16 m on the road by the time he finds rain to be falling at angle 45° .
- Q.24 The velocity displacement graph of a particle moving along a straight line is shown in figure. [4]



- (A) magnitude of acceleration of particle may decrease when it moves in +x-direction.
 (B) magnitude of acceleration of particle may increase when it moves in +x-direction.
 (C) velocity may change its direction from $x = 0$ to $x = 2$.
 (D) velocity does not change its direction from $x = 0$ to $x = 2$.

ANSWER KEY

Q.1	C	Q.2	D	Q.3	A	Q.4	A	Q.5	C	Q.6	C	Q.7	A
Q.8	D	Q.9	A	Q.10	D	Q.11	D	Q.12	C	Q.13	B	Q.14	B
Q.15	C	Q.16	B	Q.17	A,C	Q.18	B,C	Q.19	A,D	Q.20	A,B,C	Q.21	A,B
Q.22	A,C	Q.23	C,D	Q.24	A,B,C,D								